

Week 1: Jumping pegs — facilitator guide

Three tables, three adults, one move. The K–1 adult reads the student pages aloud; the two mathematicians sit with grades 2–3 and 4–5. Read this guide before the session; the section “The mathematics behind the session” is for the two mathematicians.

What the session is

The only move is the checkers-style jump: a peg hops over a peg next to it, in a straight line, onto the empty square just beyond, and the jumped peg is removed. The question at every table is the same: *an army of pegs starts on or below a line; how far past the line can it send one of its pegs?* On a strip the answer is one square, never two. On a grid, rows 1, 2, 3 and 4 can be reached with 2, 4, 8 and 20 pegs, and row 5 can never be reached, not even with infinitely many pegs (Conway’s soldiers, 1961). Every proof in the session is of one kind: find a number that no jump can increase, and show the target would need more of it than the army has. K–1 meets the phenomenon with pennies and finds the rule by exhausting the cases. Grades 2–3 meet the first structural arguments (the stranded peg, the forced last jump). Grades 4–5 meet integer Fibonacci weights that prove impossibility and preview Conway’s golden-ratio argument.

Coordinates in this guide: columns are numbered with 0 under the star, rows above the thick line are 1, 2, 3, 4 (as printed on the pages), the top row of the army is row 0, and the rows below it are -1 , -2 , -3 . A jump is written “(from) over (jumped) to (landing)” with squares as (column, row).

Session at a glance

0:00–0:05	free handling	pennies and the first page on every table; adults only demonstrate the jump when asked
0:05–0:07	launch	organizer, whole group, script below
0:07–0:50	table work	timings per table in the table sections
0:50–0:58	share	one child per table shows one thing; organizer tells the row-5 story
0:58–1:00	pack up	pennies back in the cups, pages go home

Materials and preparation

- **Pegs:** pennies, or round counters at most 2 cm across (the grades 2–5 pages say “peg”, the K–1 pages say “penny”) (cells are 2.1–2.2 cm; the 20-peg board has 2.0 cm cells). Each child needs up to 8 (K–1), 15 (grades 2–3) or 20 (grades 4–5) at once. Bring about 30 for the K–1 table, 70 for grades 2–3, 70 for grades 4–5, in three cups.
- **Pages, printed single-sided** (children put pennies on them): K–1 pages (6 sheets) $\times$ 4 copies; grades 2–3 pages (7 sheets) $\times$ 5 copies; grades 4–5 pages (7 sheets) $\times$ 4 copies; this guide $\times$ 3.
- **Pencils** and one colored pencil or crayon per child (squares get colored in, one color is enough).

- **Before the session**, each adult plays the 8-peg solution once (figure in “Solutions in pictures”). The grades 4–5 adult also walks through the 19 jumps on board F from the list, and the K–1 adult reads the K–1 section with the pages in hand.

The rules, precisely. A peg moves only by jumping: two squares in a straight line (up, down, left or right), over a peg on the square it passes, onto an empty square; the jumped peg is removed. No diagonal jumps, no jumping over empty squares, no moving without jumping. Jumps may go backward, sideways or downward. The board edge is a wall: a jump that would land off the page is not allowed. The game ends when no jump is possible, but children may stop earlier. On strips the thick bar is “the line”; on boards the thick line is between row 0 (the army’s top row) and row 1.

Whole-group launch (2 minutes)

Everyone has Problem 1 of their own pages in front of them and two pennies. Organizer: “Put a penny on each of the two gray squares. One penny may jump over the penny next to it onto the empty square right behind it. The penny that was jumped over comes off the board. Do it.” Wait until every child has jumped once. “That is the only move in today’s game. Here is the question for the whole hour: a group of pennies sits below a line. How far past the line can the group push one of its pennies? On your strips, past the thick bar. On your boards, above the thick line. Your pages say what to try. Go.” Do not explain anything else; the tables start.

K–1 table (two kindergartners and a first grader)

The adult reads each problem aloud, repeats the rules as needed, and writes numbers for children who do not write yet. Rough timings: Problems 1–4 (strips) 12 minutes; 5–6 six minutes; 7–8 ten minutes; 9–10 eight minutes; 11–12 the rest. Children work in parallel on their own pages; let them watch each other.

Problem 1. The penny lands on the third square, the first square past the bar. It is the only jump possible.

Problem 2. The only possible jump is the middle penny over the front penny onto square 4. The two pennies left (squares 1 and 4) have nobody next to them: stuck. Farthest square: the first past the bar. Pennies left: 2.

Problem 3. First the third penny over the fourth onto square 5 (the only jump); then the first over the second onto square 3; stuck with pennies on 3 and 5. Farthest: square 5, the first past the bar. Pennies left: 2.

Problem 4. Fourth over fifth onto 6; then second over third onto 4; stuck with pennies on 1, 4, 6. Farthest: square 6, the first past the bar. The star (square 7) is never reached. Pennies left: 3.

Compare, then explain (after Problem 4). Ask: “Where did the penny stop every time?” (the square right after the bar). “How many pennies were left? 1, 2, 2, 3.” Then the explanation in words a five-year-old can follow, acted out on strip 4: “When a penny jumps over the bar it leaves two empty squares behind it. The pennies behind can only ever jump one square past their own front. That is the second empty square, never the square right behind the front penny. So nobody can ever get next to the front penny again, and

the front penny never gets a second jump.” Have them check it: after the first jump on strip 4, point to the two empty squares, then let the three pennies behind do their jump and see where it lands.

Problem 5. Board A: yes, the lower penny jumps over the upper one onto the star. Board B: no; the penny stops on row 1 and both pennies left are stuck. A column of pennies is a strip standing up: one square past the front, never two.

Problem 6. Board C (three in a row): no penny can go up at all, only sideways. A penny can only jump up over a penny directly above it, and nobody has one. The highest square reached is on the starting row; children color a square on row 0. Board D (two in a column plus one to the right): row 1 only. After the upward jump the two pennies left are diagonal neighbors, not side by side, so nothing more can happen; starting with the sideways jump instead leaves two pennies that are not neighbors either.

Problem 7. Board E (the L: two in the star’s column, two more to the right on row 0): yes, in three jumps: up, then the far-right penny jumps left into the hole, then up again (figure in “Solutions in pictures”). Board F (2-by-2 square): row 1 only. Board G (T shape): row 1 only. Board H (four in a row): nothing goes up.

Problem 8. Board I (L with the arm one row lower): row 1 only. After the upward jump the sideways jump lands on the square below the hole, so the peg on row 1 never gets a partner under it. Board J (the L mirrored, star over the right-hand column): yes, the same three jumps mirrored. The two boards where a penny reached the star are E and J.

Problem 9. Board K: the L of board E, placed with its column under the star and its arm to the right. Board L (star over the third column): the mirrored L, with the arm to the left. Children often try a column of four (reaches row 1 only) or a 2-by-2 square first; let them, then ask “what did board E look like?”

Problem 10. Board M: 3 pennies cannot reach the star; the best is row 1, with two in a column. Board N: 5 pennies can, for example the L plus one penny anywhere else below the line (it never gets in the way, because the L only lands on squares of the star’s column).

Problem 11. Yes: 7 jumps, shown in “Solutions in pictures”. If they stall, say: “First make a penny stand two rows up in the star’s column, like board E. Then the pennies on the left refill the column: one jumps in along the bottom row, one jumps in along the row below, and the column climbs again.”

Problem 12. Whenever the star is reached: 7 jumps and 1 penny left. In every game from board P, jumps plus pennies left is 8, because each jump removes exactly one penny.

Hints, in order. Strips: “Which penny is able to jump? Try each one.” Board B: “Could the penny go up twice?” Boards E–J: “Before the last jump, where must the two pennies be?” (in the star’s column, on rows 1 and 0); “after the first jump up, that column has a hole; who can fill it from the side?” Board K: “Copy board E. Now board L: the star moved; which way does the arm go?”

If the table stalls. If a child jumps diagonally or over an empty square, go back to Problem 1 and replay it. If coloring is slow, the adult colors for them. If the boards confuse, turn board A on its side so the column of two looks like a strip. If the children get restless with the strips, skip Problem 4 and come back to it later.

If the table finishes early. (1) Board P mirrored: put the three second-row pennies at the right end instead of the left; it still works. (2) “Make the longest row of pennies you can on the bottom row of board P. Can any penny go up?” (never). (3) “How many pennies in one column do you need to reach the star on board A? On board B?” (2; no number of pennies in a column reaches row 2). (4) Two children race on board P and count jumps: the game is 7 jumps long whenever the star is reached.

Grades 2–3 table (four third graders)

Rough timings: Problems 1–2 eight minutes; 3–4 eight minutes; 5 five minutes; 6 eight minutes; 7 five minutes; 8–9 eight minutes; 10 as time allows. Pairs work well on Problems 7–10.

Problem 1. Strips A–D (2, 3, 4, 5 pegs starting at the left edge): the farthest square is always the first square past the bar; jumps 1, 1, 2, 2; pegs left 1, 2, 2, 3. On these strips every game is forced: the only possible jump is always at the front of the row (jumps inside the row would land on occupied squares, and the left edge is a wall). Point out what every jump does: the jumping peg lands two squares ahead of its friends and is stranded, and the row behind it is two pegs shorter.

Problem 2. Strip E (pegs on squares 3, 4, 5; two empty squares behind): the first jump is either the middle peg forward over the front one (onto square 6) or the middle peg backward over the rear one (onto square 2). Either way the two remaining pegs are not neighbors and the game is over. Farthest square: 6, the first past the bar. Extra room behind only adds a way to waste a jump.

Problem 3. Strips F–H (pegs on 2–5, one empty square behind): the possible games are: (a) 4 over 5 to 6, then 2 over 3 to 4; (b) 4 over 5 to 6, then 3 over 2 to 1; (c) 3 over 2 to 1, then 4 over 5 to 6; (d) 3 over 2 to 1, then 5 over 4 to 3. Farthest landing: square 6 in (a), (b) and (c); in (d) no peg crosses the bar at all. The star on square 7 is never reached.

Problem 4. After 4 over 5 to 6, the three pegs on 1, 2, 3 can reach square 4 (2 over 3 to 4) and then nothing. They never reach square 5. This is the whole argument in miniature: to help the stranded front peg, the pegs behind would have to reach two squares past their own front, which is the very thing we are trying to do, now with a smaller army. Say it that way; a third grader can repeat it.

Problem 5. A (column of 2): 1. B (row of 3): 0, nothing can go up because an upward jump needs a peg directly above the jumper. C (two in a column plus one beside): 1. D (T shape): 1.

Problem 6. E (the L): 2, the star is reached: up, then the far-right peg jumps left into the hole, then up. F (arm one row lower): 1. G (2-by-2): 1. H (mirrored L, star over the right column): 2, reached. I (row of 4): 0. Circle E and H.

Problem 7. There is none: every 4-peg army that reaches row 2 is the L or its mirror image. Argument, as a sequence of questions for the children. “What is the last jump?” A peg jumps up over a peg on row 1 and lands on row 2 (a sideways jump along row 2 would need two pegs already on row 2). “So just before it, what is there?” Pegs on rows 1 and 0 in the star’s column. “How did the peg on row 1 get there?” By an upward jump from row -1 over row 0 in that column (a sideways arrival along row 1 would need two pegs on row 1, which costs four pegs by itself and leaves nothing for row 0). “What did that jump leave behind?” Empty squares on rows 0 and -1 of the column. “Who can refill row 0?” Only a sideways jump along row 0, from two squares away over the neighbor, since the column below is empty. That is four pegs: two in the column, two in a row beside it on row 0, arm left or arm right, and the three jumps are forced. Any extra peg would make five.

Problem 8. The seven jumps, with pictures in “Solutions in pictures”: $(0,-1)$ over $(0,0)$ to $(0,1)$; $(2,0)$ over $(1,0)$ to $(0,0)$; $(0,0)$ over $(0,1)$ to $(0,2)$; $(-2,0)$ over $(-1,0)$ to $(0,0)$; $(-2,-1)$ over $(-1,-1)$ to $(0,-1)$; $(0,-1)$ over $(0,0)$ to $(0,1)$; $(0,1)$ over $(0,2)$ to $(0,3)$. In words: the right-hand L puts a peg on row 2; the left-hand pegs refill the column (one along row 0, one along row -1); the column climbs again and the last jump lands on the star.

Problem 9. From board K there are exactly four possible first jumps: up in column -2 , up in column -1 , up in column 0, and sideways along row -1 ($(-1,-1)$ over $(0,-1)$ to $(1,-1)$). Each of the three upward first

jumps still leads to the star; the sideways one ruins the game (row 2 is then the best). A second route to the star, starting in column -2 : $(-2,-1)$ over $(-2,0)$ to $(-2,1)$; $(0,-1)$ over $(0,0)$ to $(0,1)$; $(2,0)$ over $(1,0)$ to $(0,0)$; $(0,0)$ over $(0,1)$ to $(0,2)$; $(-1,-1)$ over $(-1,0)$ to $(-1,1)$; $(-2,1)$ over $(-1,1)$ to $(0,1)$; $(0,1)$ over $(0,2)$ to $(0,3)$: here the two left columns climb to row 1 and a sideways jump along row 1 refills the star's column. Most jumps: 7, since each jump removes one peg and the game can end with 1 peg. Pegs left when the star is reached: always 1 (games from this army end with 1, 2, 3 or 4 pegs; only the 7-jump games reach the star).

Problem 10. Nine pegs in a 3-by-3 square: row 2 (it contains an L) and not row 3; a certificate is in “The mathematics behind the session”, and the children can be told plainly: “nine pegs in a square cannot do what eight pegs in the right shape can.” All 15 squares: row 3 (the 8-peg army of board K sits inside, and the other seven pegs never get in its way) and not row 4 (fifteen pegs are never enough for row 4; the grades 4–5 table proves it with values, see below).

Hints, in order. Problem 3: “List every first jump. There are only two.” Problem 6: “Which boards contain board E, turned or not?” Problem 7: “What has to be true just before the last jump? And just before that?” Problem 8: “Board E is hiding in board K; find it; do its three jumps; now what is left, and where is the hole?” Problem 10: “Which smaller army that you know is hiding inside?”

If the table stalls. On strips, have the children say out loud which peg jumps and where it lands before moving it. On Problem 7, let them run a few failed armies and record them; the recording is the point, then walk the question sequence above. On Problem 8, give the first three jumps (they are board E) and let them find the rest.

If the table finishes early. (1) “Find every 8-peg army on board M that reaches the star on row 3.” There are exactly two: board K's army and its mirror image (five on row 0 plus three at the right end of row -1). (2) “Fill the two top rows of board M (10 pegs). How high?” Row 3, not row 4. (3) “How many jumps does a game have?” Pegs at the start minus pegs at the end, always. (4) Give them the grades 4–5 page with board F (20 pegs, star on row 4) and the list of 19 jumps from this guide.

Grades 4–5 table (two fourth graders and a fifth grader)

Rough timings: Problems 1–2 seven minutes; 3 eight minutes; 4–5 seven minutes; 6 ten minutes; 7–8 seven minutes; 9 four minutes; 10 the rest. Problem 3 and Problems 7–8 are the heart of the table: do not skip them to get to board F.

Problems 1–2. Exactly as for grades 2–3 above (strips A–D forced; strips E–F: four possible games, farthest square 6 or 5, never the star).

Problem 3. Strip G, values $0 \mid 1, 1, 2, 3 \mid 5, 8, 13$, pegs on the squares worth 1, 1, 2, 3: start 7. Forward first: the peg on 2 jumps over 3 onto 5, total $7 - 2 - 3 + 5 = 7$; then the peg on the first 1 jumps over the second 1 onto 2, total $7 - 1 - 1 + 2 = 7$; stuck (pegs on 2 and 5). Strip H, backward first: the second 1 jumps over the first 1 onto 0, total $7 - 1 - 1 + 0 = 5$; then either 2 over 3 onto 5 (total 5) or 3 over 2 onto 1 (total $5 - 3 - 2 + 1 = 1$). Star square: 8. The conclusion, to be drawn with the children at the table: every value is the sum of the two values before it ($2 = 1 + 1$, $3 = 1 + 2$, $5 = 2 + 3$, $8 = 3 + 5$, $13 = 5 + 8$), so a forward jump, which trades the two squares before a square for that square, keeps the total exactly; a backward jump trades a square and the one before it for the one two back, which is smaller, so it loses. The total starts at 7 and never goes up. A peg on the star would contribute 8 by itself. So no peg ever stands on the star, on this strip or any longer one (with n pegs the army is worth one less than the square two past the bar, by the identity $F_1 + \cdots + F_n = F_{n+2} - 1$). This is a proof with no cases to check, and

the same idea runs the rest of the table.

Problem 4. A: 2 pegs, a column of two under the star. B: 4 pegs, the L (two in the column under the star, two beside on row 0, arm left or right): up, in along row 0, up. Nothing smaller works (Problem 5).

Problem 5. No 3-peg army reaches row 2. Column of 3: row 1 (a column is a strip on end). Two in a column plus a neighbor: row 1. Row of 3: nothing goes up. Three pegs not touching: nothing moves. The argument: the last jump lands on row 2 from row 0 over row 1 in one column; the row-1 peg got there by a jump from row -1 over row 0 in that column, which emptied row 0; refilling row 0 takes two more pegs in a row beside it. That is four.

Problem 6. Eight pegs: five across row 0 under the star and three on row -1 at the left end (columns -2 , -1 , 0), or the mirror image (three at the right end). On this board those are the only two 8-peg armies that work. A third 8-peg army exists but needs four rows below the line: five across row 0 and three straight down under the star (a T). Seven pegs never suffice, which is Problems 7–8. Children who start from the 3-by-3 square of nine reach row 2 only; children who fill all 15 squares reach row 3 but have no idea why; steer both to the hints.

Problem 7. With the 8-peg army on board D: $5 + 3 + 3 + 2 + 2 + 3 + 2 + 1 = 21$ at the start, and 21 after every one of the seven jumps: each jump lands on a square from the two squares beyond it in a straight line away from the star, and each value on board D is the sum of the two values beyond it in any straight line away from the star ($5 = 3 + 2$ along the row and $5 = 3 + 2$ down the column, $8 = 5 + 3$, $13 = 8 + 5$, $21 = 13 + 8$, $3 = 2 + 1$, $2 = 1 + 1$). Sideways jumps lose: for example two pegs on $(-1, 0)$ and $(0, 0)$, worth $3 + 5$, and the left one jumps over the other onto $(1, 0)$, worth 3: the total drops by 5, the value of the jumped peg. Downward jumps lose more: $(0, 0)$ over $(0, -1)$ onto $(0, -2)$ turns $5 + 3$ into 2. So no jump ever raises the total.

Problem 8. The seven largest values below the line are 5, 3, 3, 3, 2, 2, 2, sum 20; the star is worth 21. The proof that seven pegs can never reach the star, said at the table: “Any seven squares below the line are worth at most 20. No jump raises the total. A peg on the star would make the total at least 21.” This also explains Problem 6: the 8-peg army is worth exactly 21, so in a solution every jump must keep the total, which is why all seven jumps go toward the star and why there are so few solutions.

Problem 9. Highest row: 3 (the 8-peg army sits inside the 15 and the other seven pegs never block it). The 15 squares are worth $15 + 9 + 6 = 30$; the star on row 4 is worth 34. So no army on this board reaches row 4, however it plays. Board E is board D with one more row because the star moved up one row: every square is one step farther from the star, so every value shifts one place along 1, 1, 2, 3, 5, 8, 13, 21, 34.

Problem 10. The nineteen jumps on board F, in the page’s notation (column letter, row number from the bottom; the star is e8): 1. g3–g5, 2. f3–f5, 3. g5–e5, 4. e4–e6, 5. c4–e4, 6. a4–c4, 7. e3–e5, 8. e5–e7, 9. c3–c5, 10. d2–d4, 11. b2–d2, 12. e1–e3, 13. g2–e2, 14. e2–e4, 15. d1–d3, 16. d3–d5, 17. c5–e5, 18. e4–e6, 19. e6–e8. Pictures are in “Solutions in pictures”. Structure to point out: jumps 1–3 put a peg on row 1 of the star’s column from the side (the two columns on the right climb to row 1, then one peg jumps sideways along row 1), and jump 4 climbs to row 2; jumps 5–8 refill $(0, 0)$ from the left, bring the spare left pegs closer, and climb to row 3; jumps 9–18 rebuild the column again, from below (jumps 12–14) and by a second sideways jump along row 1 (jumps 9–11, 15–17), and jump 18 climbs to row 2; jump 19 lands on the star. Twenty is the smallest army that reaches row 4 (Winning Ways; see the mathematics section). The pattern 2, 4, 8 does not continue to 16.

Hints, in order. Problem 3: “Write the value of each square under a peg before you add.” Problem 6: “Where are the two pegs just before the last jump?”; “you know a 4-peg army that puts a peg on row 2, build it under the star”; “after it has worked, the column below is empty: refill square $(0, 0)$ along row 0

from the left, then square $(0, -1)$ along row -1 from the left, then climb.” Problem 8: “If you could choose any seven squares, which ones, and what are they worth?” Problem 10: “Find the 8-peg army inside board F, but do not use it first: the first three jumps are on the right.”

If the table stalls. Problem 3: do the arithmetic together, one jump at a time, on one strip. Problem 6: give the L hint at once; if they want to try the 3-by-3 square, let them (row 2 only), then say that eight pegs in the right shape beat nine in a square. Problem 10: give the first eight jumps from the list and let them find the rest; a peg on row 3 with the column empty below is the natural checkpoint.

If the table finishes early. (1) “Find the third 8-peg army” (the T; it needs four rows below the line, so use the lower part of board F with a star drawn three rows above a chosen row). (2) “Guess the smallest army for row 4 from 2, 4, 8.” Then: 20, and board F is the proof that 20 is enough. (3) “Is there a row that can never be reached, with as many pegs as you like?” Row 5; the story is told at the share. (4) “On board D, how many 8-peg armies are worth exactly 21, and do they all reach the star?” There are 63 (the square worth 5, the three worth 3, three of the five worth 2 and one of the six worth 1), and only the two solutions reach the star: enough value is necessary, not sufficient.

Closing share (8 minutes)

One child from each table shows one thing with the pennies while the others watch. K-1: strip 4 (“the penny always stops right after the bar”). Grades 2-3: board K, the 8 pegs reaching row 3 (one child plays, one counts the jumps), and board B, the row that cannot rise. Grades 4-5: strip G, “worth 7, the star is worth 8”; board D, “the seven best squares are worth 20, the star is worth 21”. Then the organizer, two minutes: “Rows 1, 2, 3, 4 need 2, 4, 8 and 20 pegs. Now imagine the army fills everything below the line, forever, in every direction. Row 5 can still never be reached. John Conway proved it in 1961, with values exactly like the ones on board D: each square is worth the sum of the two beyond it, so a jump toward the star never gains and every other jump loses. His values are not whole numbers; they are powers of the golden ratio, and with them the whole infinite army is worth exactly one star.” Hold up board E and read the values up the star’s column: 2, 3, 5, 8, 13, 21, 34.

The mathematics behind the session

Monovariants. Every impossibility argument of the session has the shape: assign each square a value, check that no legal jump increases the total value of the pegs on the board, and compare the army’s starting total with the value a peg on the target would contribute by itself. Values must be nonnegative so that the other pegs cannot compensate. Such a function is called a pagoda function in *Winning Ways*; the general idea is the monovariant (or potential function) of combinatorics courses and olympiad problems.

The strip, three ways. Theorem: a finite army on a strip never puts a peg more than one square beyond its initial front. (i) Stranded pegs (grades 2-3). Starting from a solid row, the only possible jumps are at the two ends of the row: an interior jump would land on an occupied square. An end jump strands the jumper beyond a gap of two empty squares; the next end jump on the same side lands in the near one of those two squares; so the stranded pegs sit on alternate squares separated by single empty squares, the row shrinks by two per jump, and a square between two stranded pegs can never be filled because filling a square needs two adjacent pegs right next to it, and the only adjacent pegs are in the row, whose jumps land exactly one square past its end. Consequences: a row of n pegs allows exactly $\lfloor n/2 \rfloor$ jumps,

leaves $\lceil n/2 \rceil$ pegs, and sends a peg exactly one square past its front. (ii) Descent, valid for any starting arrangement, by induction on the number of pegs. Let F be the front. The first jump beyond F lands on $F + 1$ (from $F - 1$ over F) and leaves F and $F - 1$ empty, so the other pegs are at or behind $F - 2$. The lone peg on $F + 1$ can take part in a jump only once square F is occupied (square $F + 2$ can be reached only from F over $F + 1$). The remaining pegs form a smaller army whose front is at most $F - 2$, and by the induction hypothesis they never reach F . (iii) Fibonacci weights (grades 4–5). Give the squares the values $F_1, F_2, F_3, \dots$ from the back of the army forward (so the army occupies $F_1, \dots, F_n$, the square past the front is F_{n+1} , and the square two past it is F_{n+2}); with room behind, prepend $F_0 = 0$, which still satisfies the recurrence. A forward jump replaces $F_k + F_{k+1}$ by F_{k+2} and keeps the total; a backward jump replaces $F_{k+1} + F_{k+2}$ by F_k and loses $2F_{k+1}$. The army is worth $F_1 + \dots + F_n = F_{n+2} - 1$, one less than the square two past its front. Strip G is the case $n = 4$: $7 < 8$.

Conway's soldiers. On the infinite grid with the army filling the half-plane of rows ≤ 0 , give the square at Manhattan distance d from the target the value φ^{-d} , where $\varphi = (1 + \sqrt{5})/2$ and $\varphi^{-1} + \varphi^{-2} = 1$. A jump toward the target replaces $\varphi^{-(d+1)} + \varphi^{-(d+2)}$ by φ^{-d} : no change. A sideways jump (lander and jumper at the same distance) loses the jumped peg's value, and a jump away loses more. For the target on row 5, row 0 is worth $\varphi^{-5}(1 + 2\varphi^{-1} + 2\varphi^{-2} + \dots) = \varphi^{-5}(1 + 2\varphi) = \varphi^{-5}\varphi^3 = \varphi^{-2}$, each lower row is worth φ^{-1} times the one above, and the whole army is worth $\varphi^{-2}/(1 - \varphi^{-1}) = 1$, exactly the target. A finite sequence of jumps moves finitely many pegs, which are worth strictly less than 1, and the total never increases, so no peg ever stands on row 5. The golden ratio is forced: a ratio r with $r + r^2 < 1$ would let toward-jumps gain, and a ratio with $r + r^2 > 1$ makes the army worth more than the target. The 1-D theorem is the same computation one dimension down: the half-line is worth exactly the square two past the front.

Minimal armies, 2, 4, 8, 20. Row 1 needs 2 pegs, row 2 needs 4, row 3 needs 8, row 4 needs 20 (Winning Ways). The weights explain most of it. For a target on row 2 the two heaviest cells below the line weigh $\varphi^{-2} + \varphi^{-3} = \varphi^{-1} < 1$ and the three heaviest $\varphi^{-1} + \varphi^{-3} < 1$; the L weighs $\varphi^{-2} + 2\varphi^{-3} + \varphi^{-4} = 1$ exactly. For row 3 the seven heaviest cells weigh $\varphi^{-3} + 3\varphi^{-4} + 3\varphi^{-5} \approx 0.944 < 1$ while each of the three 8-peg armies (the T and the two “five plus three” shapes) weighs exactly 1. These are the only 8-peg armies that reach row 3 and the L and its mirror are the only 4-peg armies that reach row 2 (exhaustive search, and for 4 pegs the argument in the grades 2–3 section). An army worth exactly the target can only win by making toward-jumps throughout and ending with the target peg alone, which is why the solutions are so rigid. For row 4 the sixteen cells at distance at most 7 weigh ≈ 0.936 , the nineteen heaviest weigh exactly 1, and the twenty heaviest weigh ≈ 1.021 ; a direct check shows no 20-peg army inside the 25 cells of distance at most 8 works, so a minimal army must reach out to distance 9, and the army on board F (weight exactly 1, nineteen toward-jumps) is one of a handful of 20-peg solutions. Children's guess “16” from 2, 4, 8 is exactly the kind of pattern this session likes to break.

Integer weights on finite boards. Boards D and E use F_{N-d} with $N = 8$ and 9 and d the Manhattan distance to the star. Because the Fibonacci recurrence holds along every straight line leading away from the star, these are valid pagoda functions on the whole printed board (the edge needs no care as long as N is at least the largest distance, so that all values are nonnegative), and the comparison is clean: the seven heaviest army squares are worth $F_{N-3} + 3F_{N-4} + 3F_{N-5} = F_N - F_{N-6}$, short of the star by F_{N-6} , which is 1 on board D. The full 15-peg board is worth 30 against a row-4 star worth 34; with Conway's weights any 15 cells weigh at most 0.902, so 15 pegs never reach row 4 on any board. The strip values $0 \mid 1, 1, 2, 3 \mid 5, 8, 13$ are the one-dimensional case.

Pagoda functions as certificates. Any nonnegative function p on squares with $p(\text{landing}) \leq p(\text{jumper}) + p(\text{jumped})$ for every possible jump certifies that a target t is unreachable from an army A whenever $\sum_{a \in A} p(a) < p(t)$; finding the best such certificate is a linear program, and small integer ones often

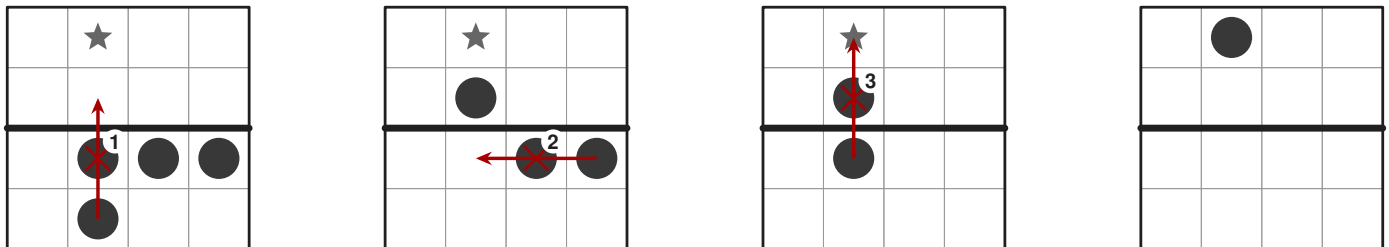
exist. Two for the 3-by-3 block of nine on board M (grades 2–3, Problem 10): put 0, 1, 0, 1, 1, 2, 0 on rows $-2, -1, 0, 1, 2, 3, 4$ of the three even columns $(-2, 0, 2)$ and 0 on the odd columns; every jump satisfies the inequality, the block is worth 1 and each row-3 square in an even column is worth 2. For the odd columns put 0, 1, 1, 2, 3, 5, 0 on rows $-2, \dots, 4$ of columns -1 and 1 and 0 elsewhere; the block is worth 4 and the row-3 squares in odd columns are worth 5. Together they show the block never reaches row 3; it reaches row 2 because it contains an L. (Found with a 35-variable integer program; checking them by hand takes a minute.) Pagoda functions are not complete certificates: the nineteen heaviest cells for row 4 weigh exactly 1 and still fail, for combinatorial reasons.

Where it leads. Potential functions and monovariants (termination arguments, Lyapunov functions, the “semi-invariant” problems of olympiad combinatorics); peg solitaire on graphs, where pagoda functions and linear-programming duality are the main tools (Beeler and Hoilman, 2011); Conway’s army in d dimensions, where the reachable limit is row $3d - 2$ (Eriksson and Lindström, 1995: 1 on the line, 4 in the plane, 7 in space); and row 5 itself, which can be reached in the plane if infinitely many jumps are allowed (Tatham and Taylor), so the finiteness in Conway’s proof is doing real work. Reading: Berlekamp, Conway and Guy, *Winning Ways for Your Mathematical Plays*, vol. 4, 2nd ed., chapter “Purging Pegs Properly”, section on the Solitaire Army; M. Aigner, “Moving into the desert with Fibonacci”, *Mathematics Magazine* 70 (1997); R. Beeler and D. Hoilman, “Peg solitaire on graphs”, *Discrete Mathematics* 311 (2011); H. Eriksson and B. Lindström, “Twin jumping checkers in $\mathbb{Z}^d$ ”, *European Journal of Combinatorics* 16 (1995).

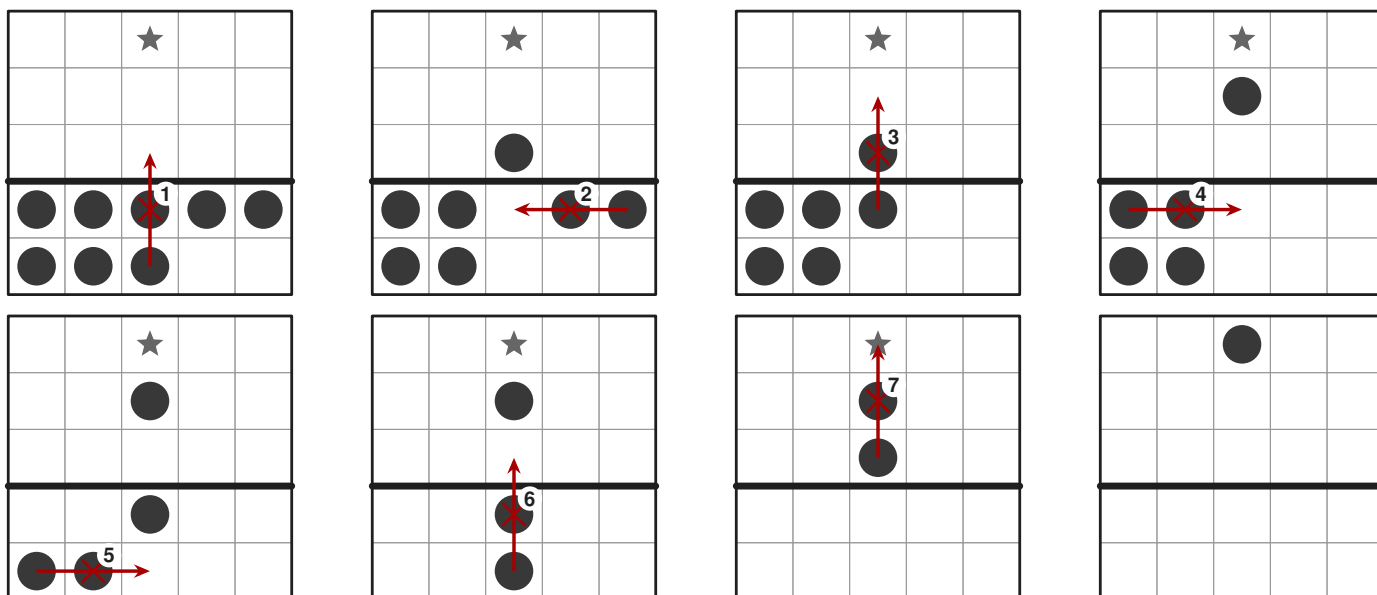
Solutions in pictures

Each picture shows the position before a jump; the arrow is the jump, the cross marks the peg that is removed, and the last picture is the final position.

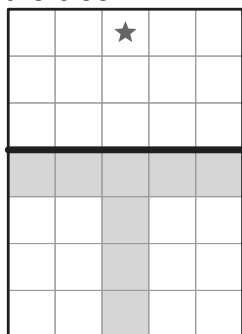
The L (boards E, J, K, L of K–1; E, H of grades 2–3; B of grades 4–5).



The 8-peg army (board P of K–1; boards K, L of grades 2–3; board C of grades 4–5).



The T-shaped 8-peg army and the two pagoda certificates for the 3-by-3 block. Left: the third 8-peg army, which needs four rows below the line. Middle and right: the two certificates (values shown in the squares, blank means 0; the targets are the row-3 squares of the valued columns); the dark squares are the block.



2		2		2
1		1		1
1		1		1
1		1		1

	5		5	
	3		3	
	2		2	
	1		1	
	1		1	

The 20-peg army (board F of grades 4–5): nineteen jumps. In the page's notation: 1. g3–g5, 2. f3–f5, 3. g5–e5, 4. e4–e6, 5. c4–e4, 6. a4–c4, 7. e3–e5, 8. e5–e7, 9. c3–c5, 10. d2–d4, 11. b2–d2, 12. e1–e3, 13. g2–e2, 14. e2–e4, 15. d1–d3, 16. d3–d5, 17. c5–e5, 18. e4–e6, 19. e6–e8.

